

Active Directory Home Lab

Technical Documentation & Walkthrough

May 2026

Lab Environment

This lab was built in VMware Workstation Pro to simulate a real enterprise Active Directory environment for hands-on helpdesk training.

Lab Configuration	
Hypervisor	VMware Workstation Pro
Server OS	Windows Server 2019
Domain Name	lab.local
Roles Installed	AD DS (Active Directory Domain Services), DNS
Client OS	Windows 10/11 (domain-joined workstation)
Tools Used	Active Directory Users and Computers (ADUC), Server Manager

Objectives

This lab covers the core Active Directory tasks performed daily by Tier 1 helpdesk technicians:

- Deploy and configure a Windows Server domain controller
- Create and organize Organizational Units (OUs) to mirror a company structure
- Create and manage user accounts
- Create security groups and assign user memberships
- Perform common account management tasks: password resets, account unlocks, and disabling accounts

Section 1: Domain Controller Setup

Server Manager — Roles Installed

The domain controller was configured with two critical roles: Active Directory Domain Services (AD DS) and DNS. Both roles show green Manageability status in Server Manager, confirming the domain is healthy and operational.

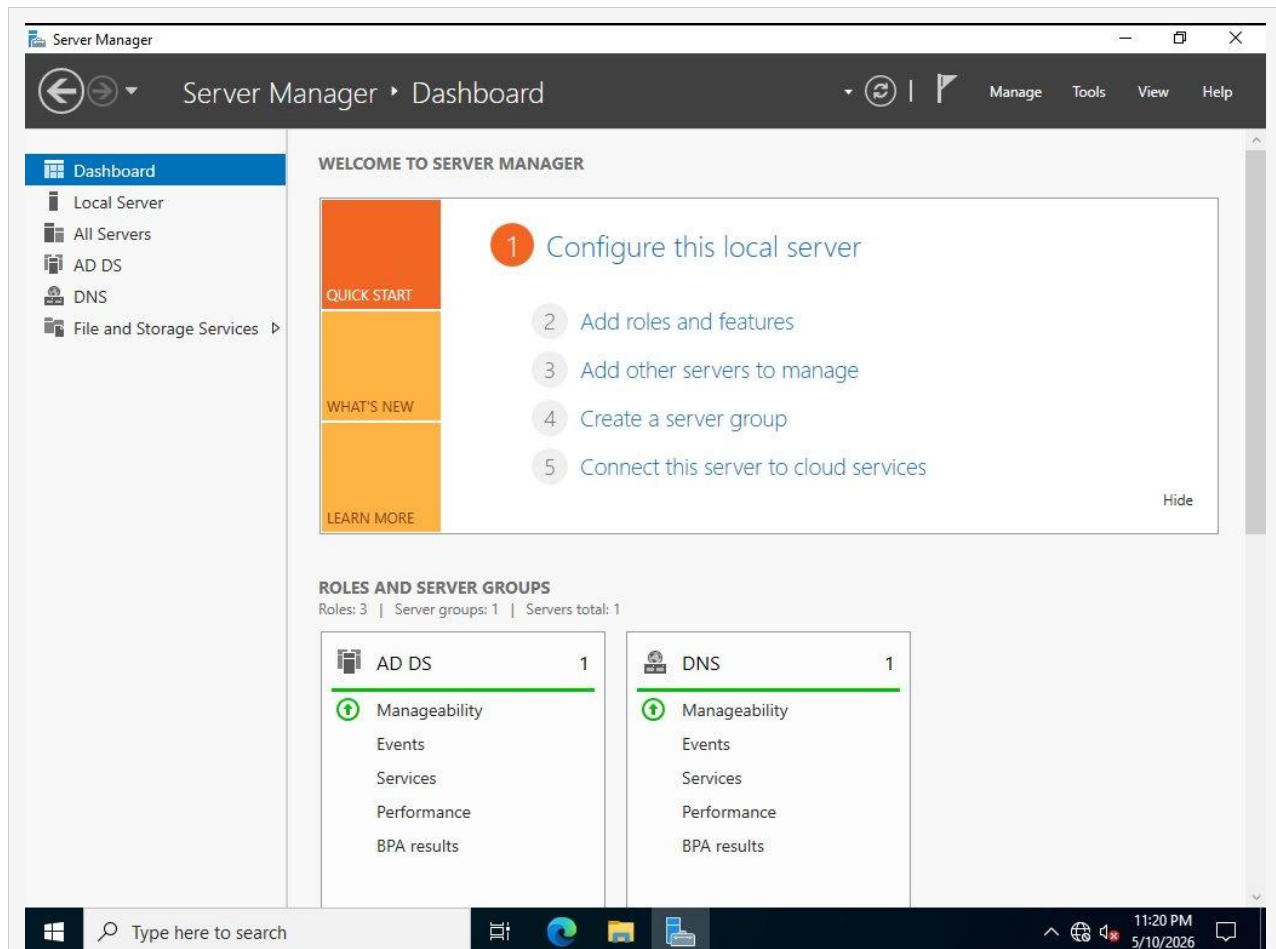


Figure 1 — Server Manager Dashboard showing AD DS and DNS roles running on lab.local

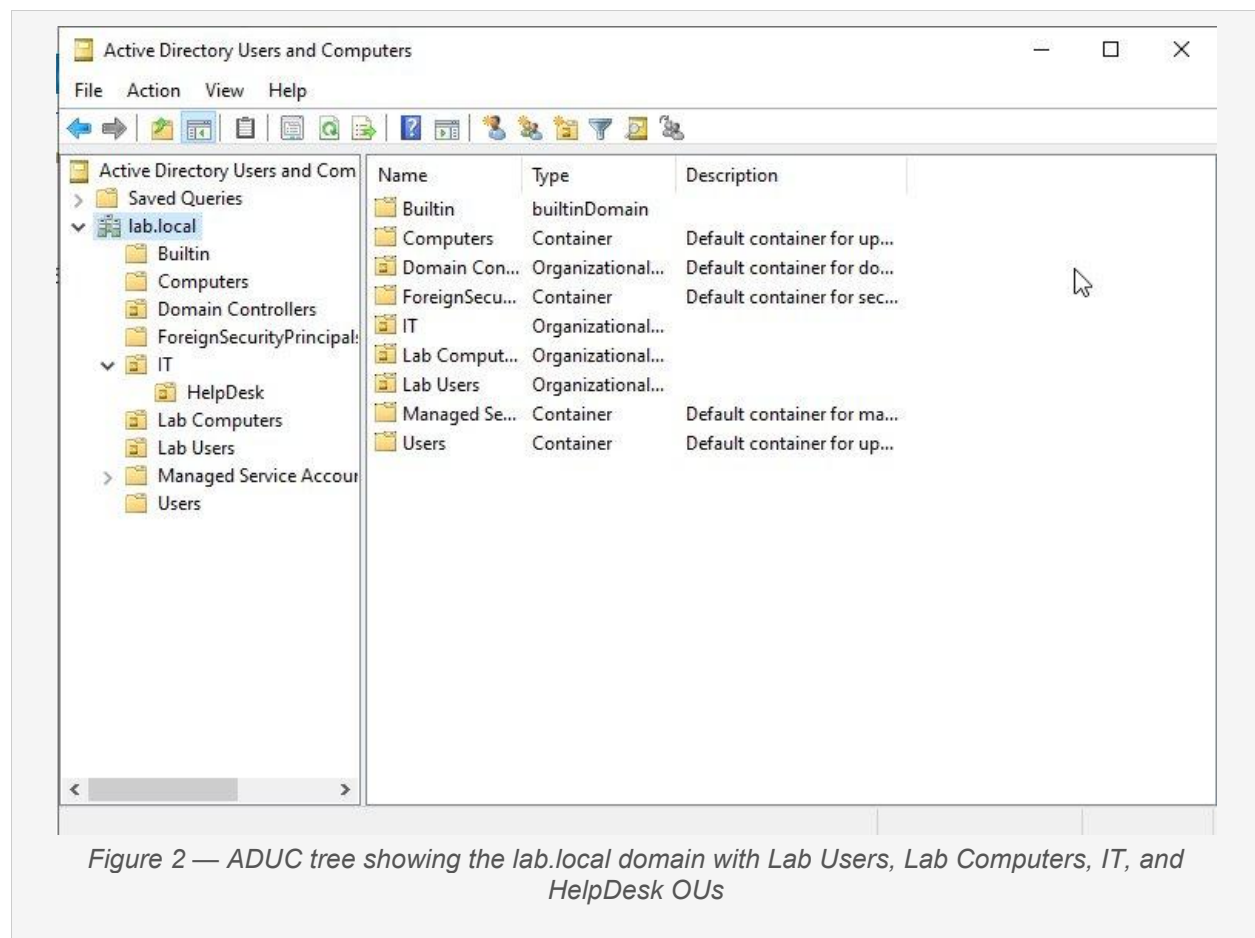
Section 2: Organizational Unit Structure

Creating OUs in Active Directory Users and Computers

Organizational Units (OUs) act as folders inside Active Directory that organize users, computers, and groups. They also allow Group Policy to be applied at a granular level. The following OU structure was created to simulate a small company:

- Lab Users — contains all standard domain user accounts
- Lab Computers — contains all workstation computer objects
- IT — represents the IT department
- IT > HelpDesk — sub-OU for helpdesk technician accounts

OUs were created by right-clicking the domain (lab.local) in ADUC, selecting New > Organizational Unit, and providing a name. The "Protect container from accidental deletion" option was left enabled as a best practice.



Section 3: User Account Management

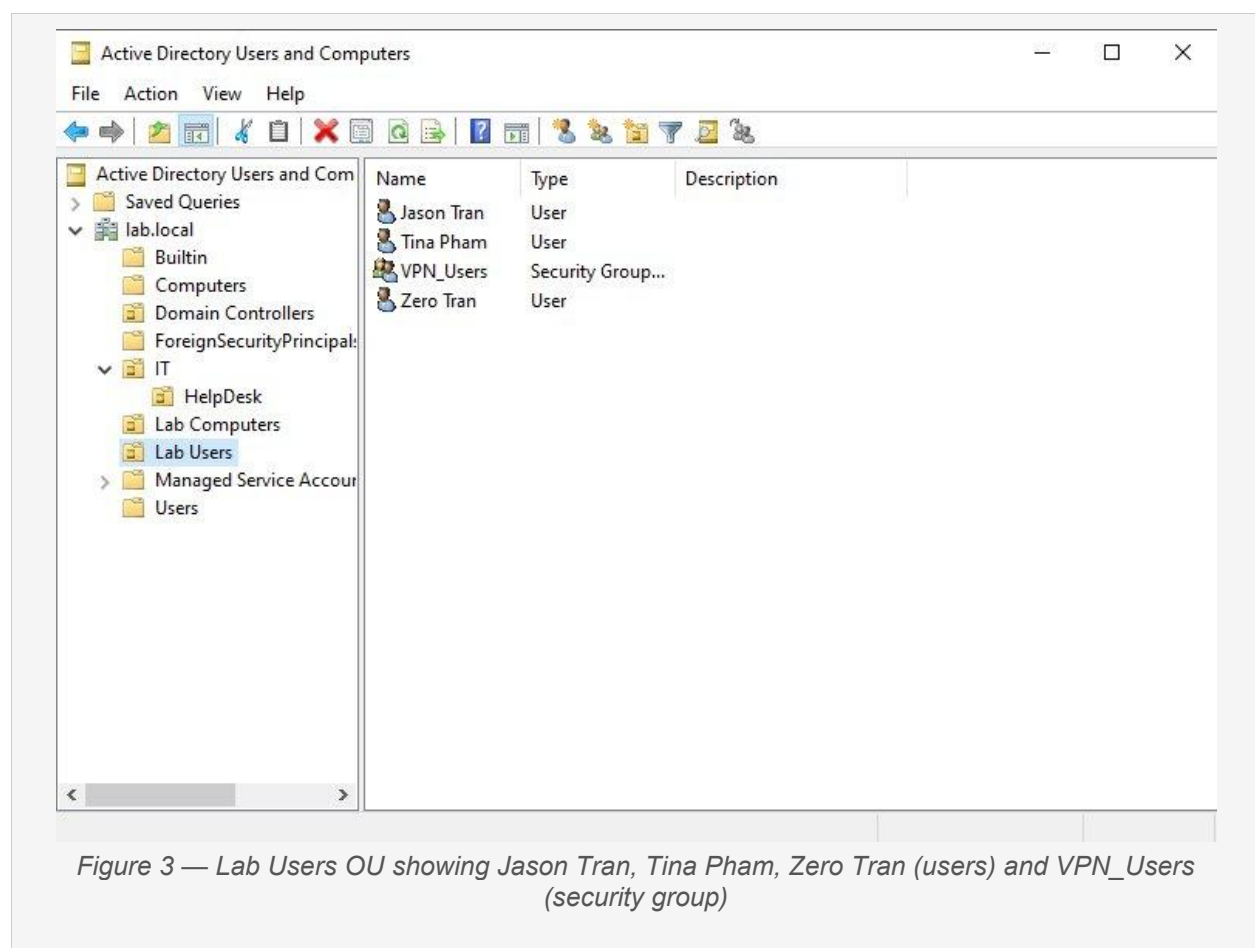
Creating Domain User Accounts

Three user accounts were created inside the Lab Users OU to simulate employees onboarding to the domain. Users were created by right-clicking Lab Users > New > User

and filling in the first name, last name, and user logon name (UPN format: username@lab.local).

The following account options were configured at creation, matching real-world helpdesk standards:

- User must change password at next logon — enforces a password reset on first login
- Account expiration set to Never for standard permanent employees



Section 4: Security Group Management

Creating Security Groups and Assigning Members

A security group named VPN_Users was created inside the Lab Users OU. Security groups are the standard mechanism in Active Directory for controlling access to resources such as VPN connections, file shares, and applications.

The group was configured as:

- Group scope: Global — accessible across the domain
- Group type: Security — used to assign permissions (not a distribution/email list)

Jason Tran and Tina Pham were added as members via the Members tab in group Properties. In a real environment, adding a user to VPN_Users would grant them VPN access without requiring individual permission changes.

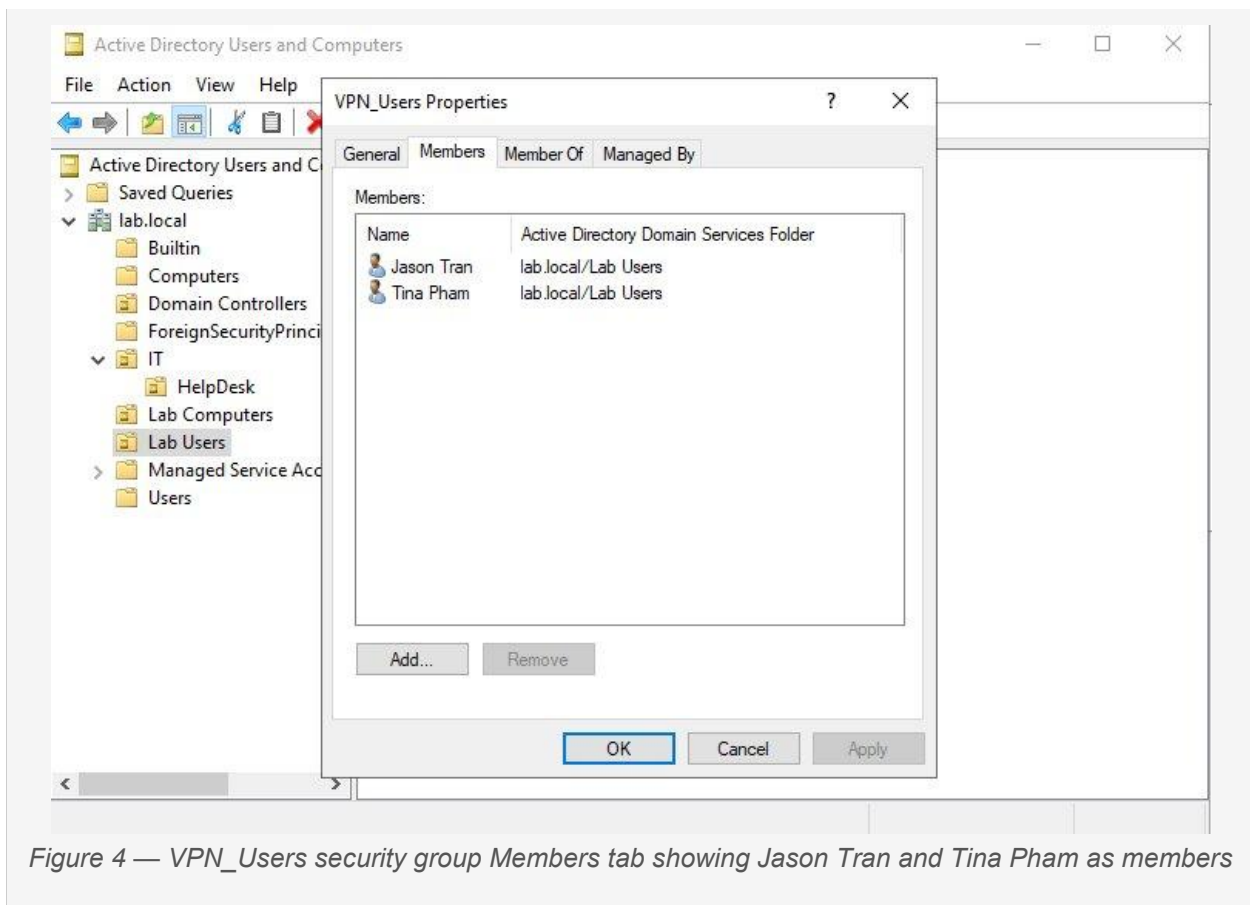


Figure 4 — VPN_Users security group Members tab showing Jason Tran and Tina Pham as members

Section 5: Account Properties and Options

Viewing and Configuring User Account Settings

The Account tab in user Properties is one of the most important screens for a Tier 1 helpdesk technician. It provides access to:

- User logon name (UPN) — the username used to log into the domain
- Pre-Windows 2000 logon name (DOMAIN\username format)
- Unlock account checkbox — used when a user is locked out after too many failed password attempts
- Account options — including "User must change password at next logon" and "Password never expires"
- Account expiration — for contractors or temporary employees

In this screenshot, Jason Tran's account is configured with "User must change password at next logon" checked, which is standard for newly created or reset accounts.

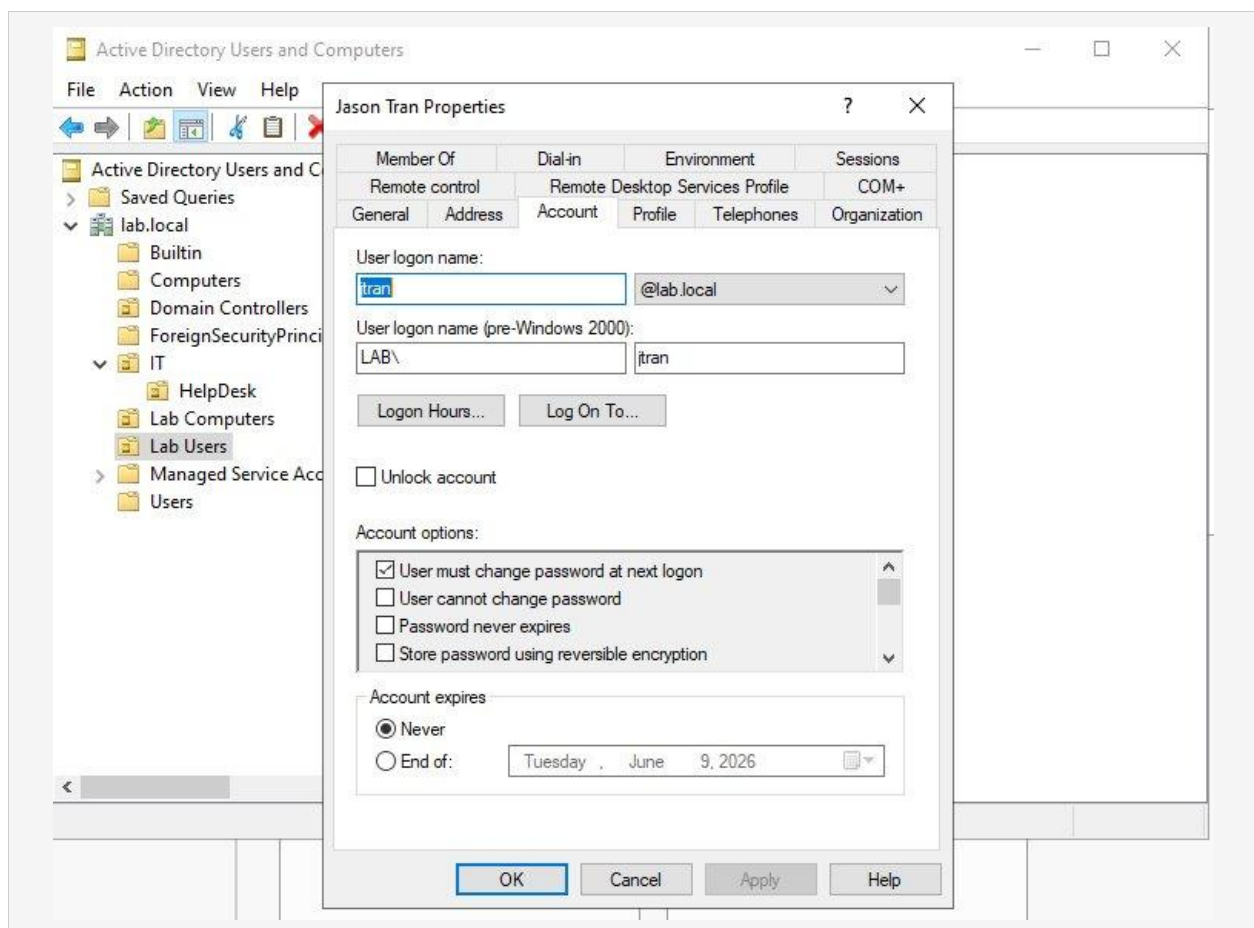


Figure 5 — Jason Tran's Account tab showing logon name (tran@lab.local), account options, and expiration settings

Section 6: Password Reset

Resetting a Domain User Password

Password resets are one of the most common Tier 1 helpdesk tickets. The process in Active Directory is:

- Right-click the user in ADUC
- Select Reset Password
- Enter and confirm the new temporary password
- Check "User must change password at next logon" so the user sets their own password immediately
- If the account is locked out, check "Unlock the user's account" in the same dialog

The Reset Password dialog also displays the current Account Lockout Status on the domain controller, which allows a technician to confirm whether an unlock is needed in the same step.

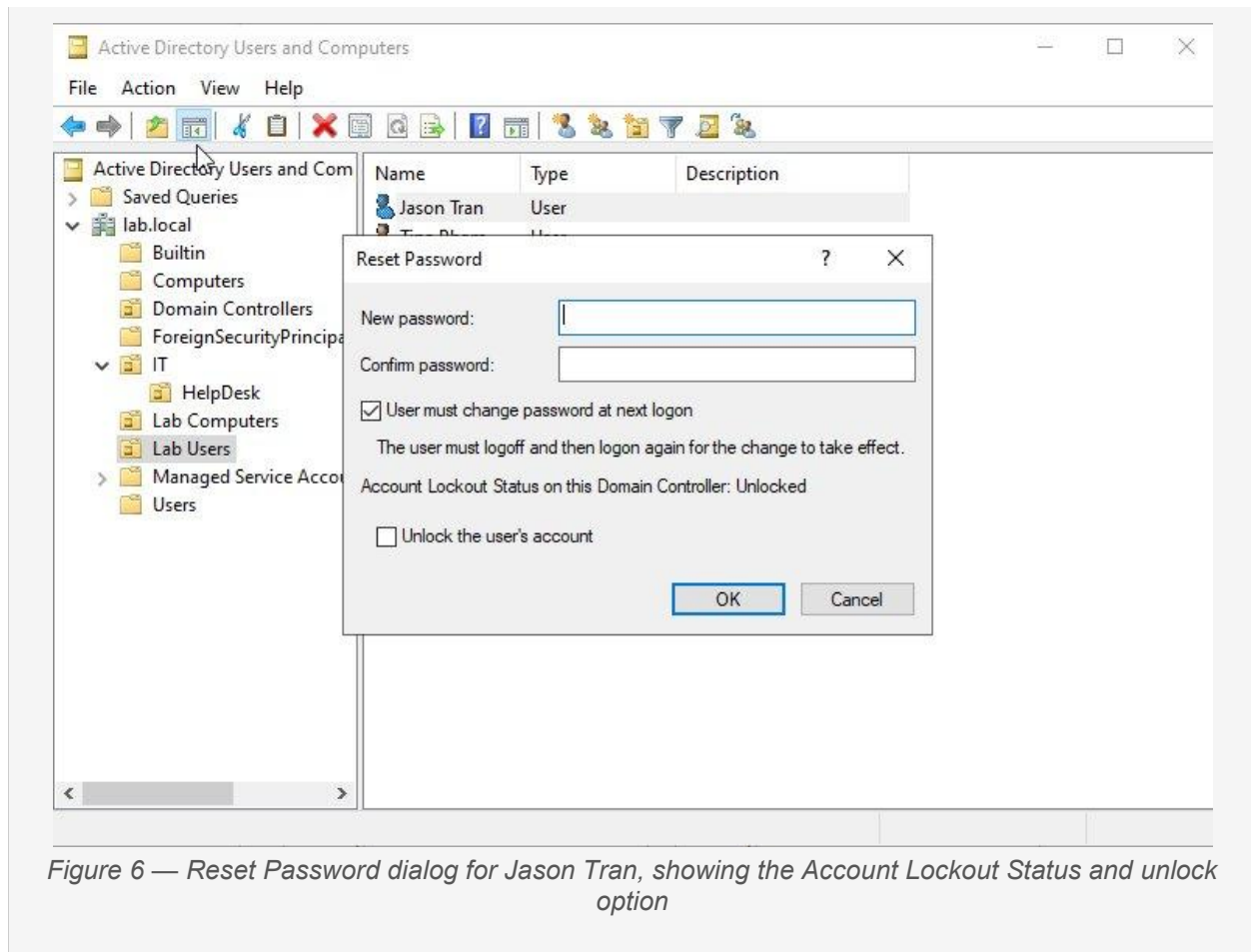


Figure 6 — Reset Password dialog for Jason Tran, showing the Account Lockout Status and unlock option

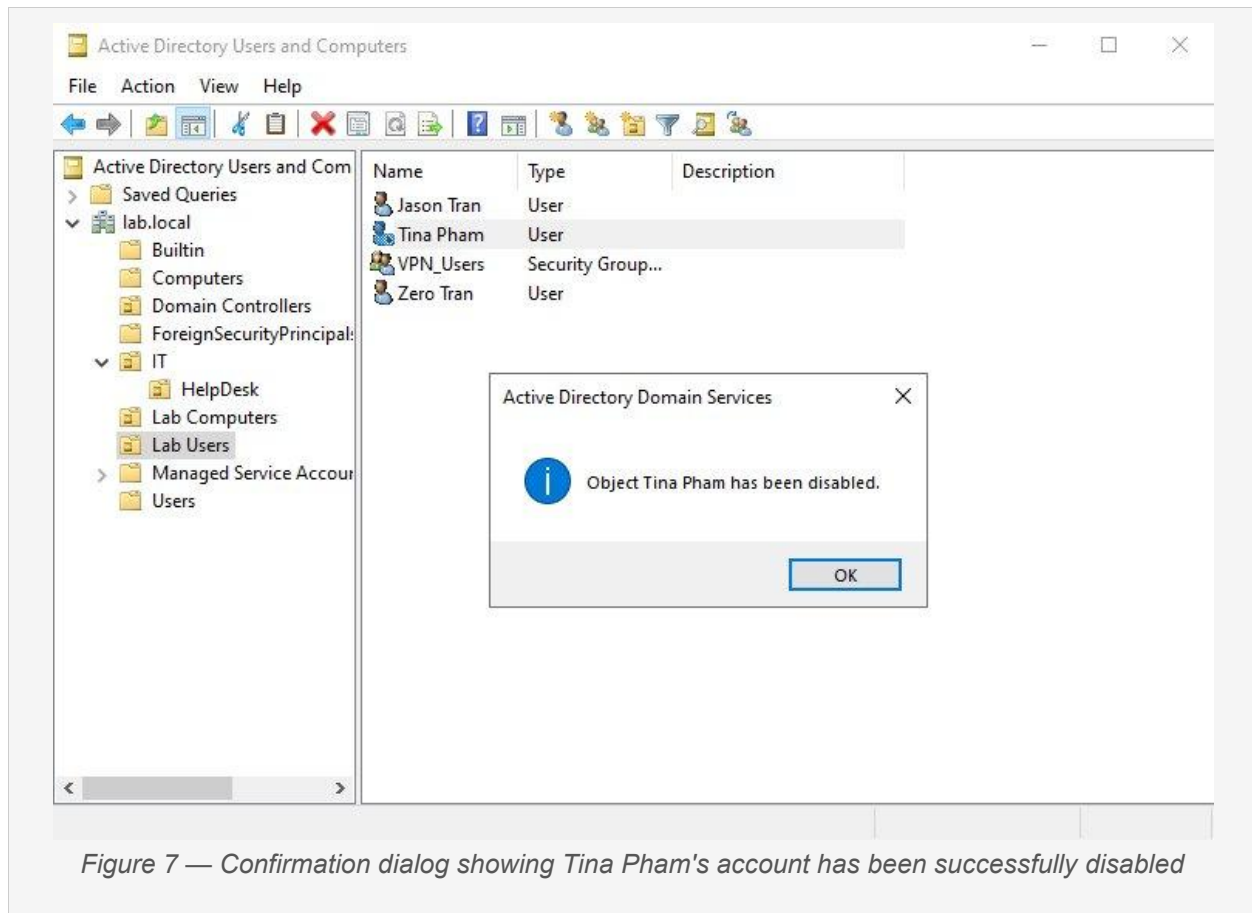
Section 7: Disabling a User Account

Disabling Accounts for Offboarding or Leave

When an employee leaves the company or goes on extended leave, their account is disabled rather than immediately deleted. This preserves their data, group memberships, and permissions while preventing login. The process is:

- Right-click the user in ADUC
- Select Disable Account
- A confirmation dialog confirms the account has been disabled
- The user icon in ADUC displays a downward arrow indicating disabled status

Tina Pham's account was disabled as a demonstration. The Active Directory Domain Services confirmation popup confirms "Object Tina Pham has been disabled." To re-enable, right-click the user and select Enable Account.



Section 8: Group Policy Object (GPO)

GPO: Password Policy — Linked to lab.local

A GPO named Password Policy was created and linked directly to the lab.local domain, meaning it applies to all users and computers in the domain. It was created by right-clicking lab.local in GPMC and selecting "Create a GPO in this domain, and Link it here."

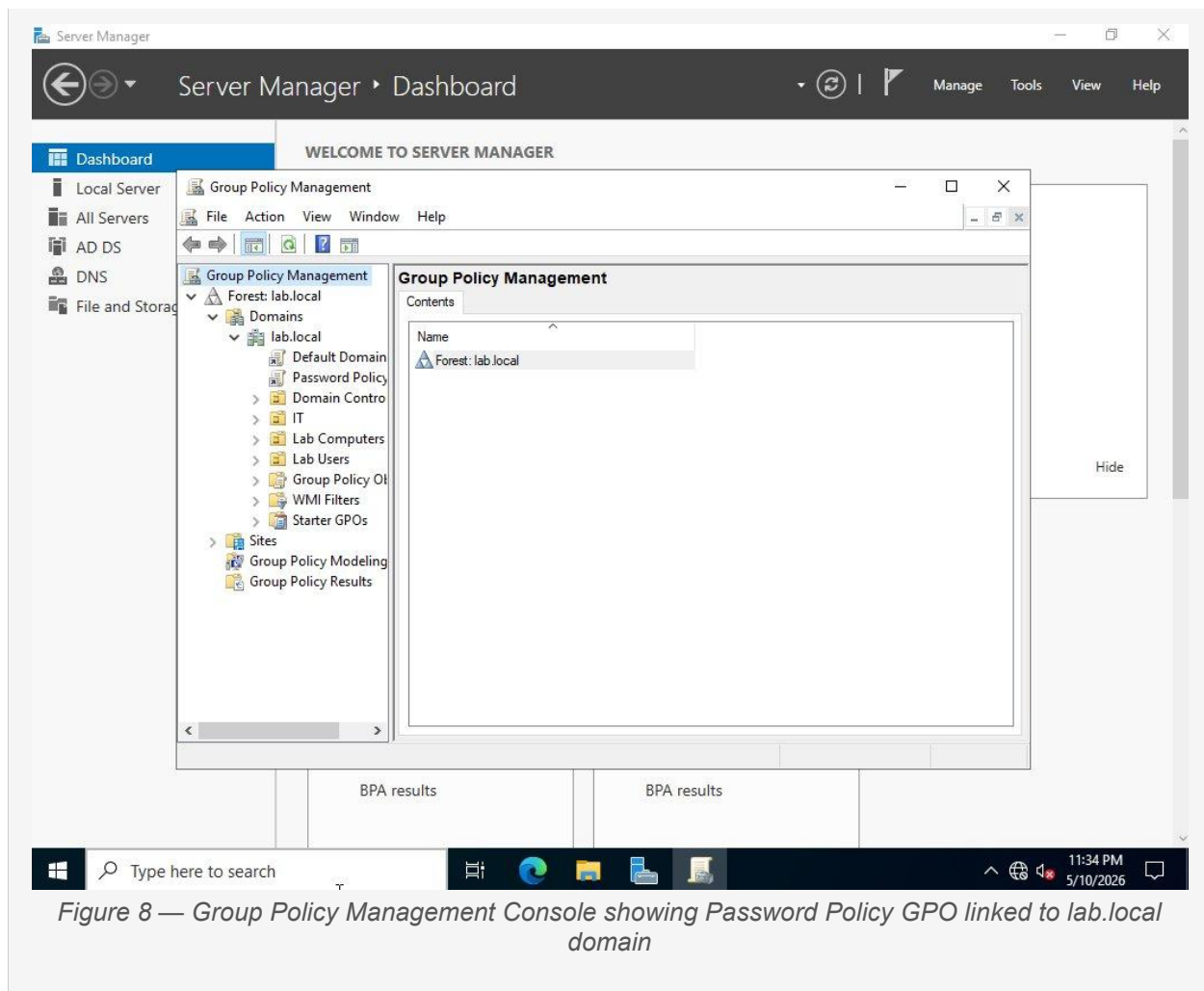


Figure 8 — Group Policy Management Console showing Password Policy GPO linked to lab.local domain

Password Policy Settings

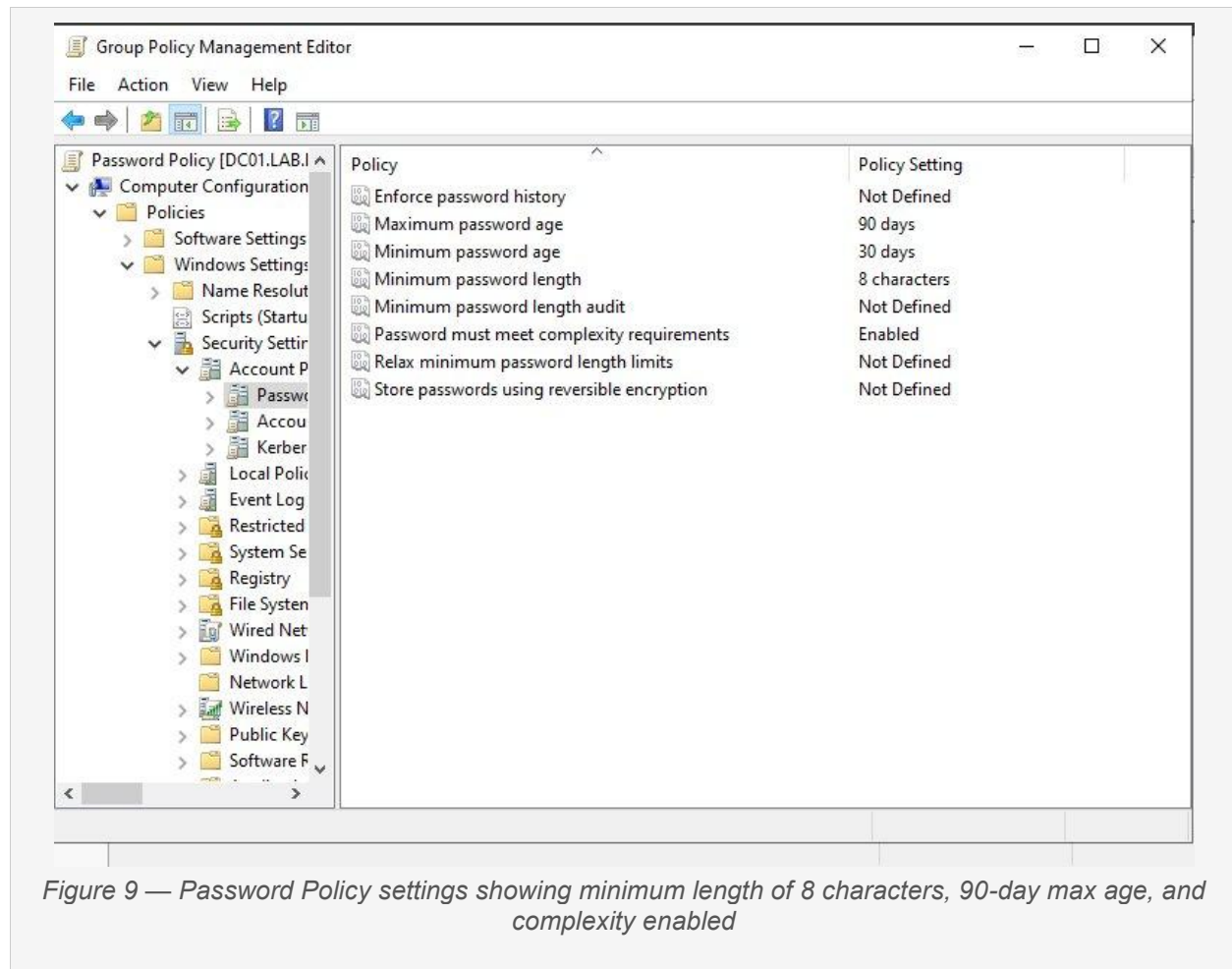
The GPO was edited by right-clicking it and selecting Edit, then navigating to:

Computer Configuration → Policies → Windows Settings → Security Settings → Account Policies → Password Policy

The following settings were configured:

Policy Setting	Value Configured	Purpose
Maximum password age	90 days	Forces password change every 90 days

Minimum password age	30 days	Prevents users changing password immediately to reuse old one
Minimum password length	8 characters	Requires at least 8 characters
Password must meet complexity requirements	Enabled	Requires uppercase, lowercase, number, or symbol



Account Lockout Policy Settings

Account lockout settings are configured in the same GPO under Account Lockout Policy. These settings are directly relevant to Tier 1 helpdesk work — account lockouts are one of the most common support tickets in any organization.

Policy Setting	Value Configured	Purpose
Account lockout threshold	5 invalid logon attempts	Locks account after 5 wrong passwords
Account lockout duration	30 minutes	Account auto-unlocks after 30 min, or manually by helpdesk
Reset account lockout counter after	30 minutes	Resets the bad-attempt counter after 30 min of no failures

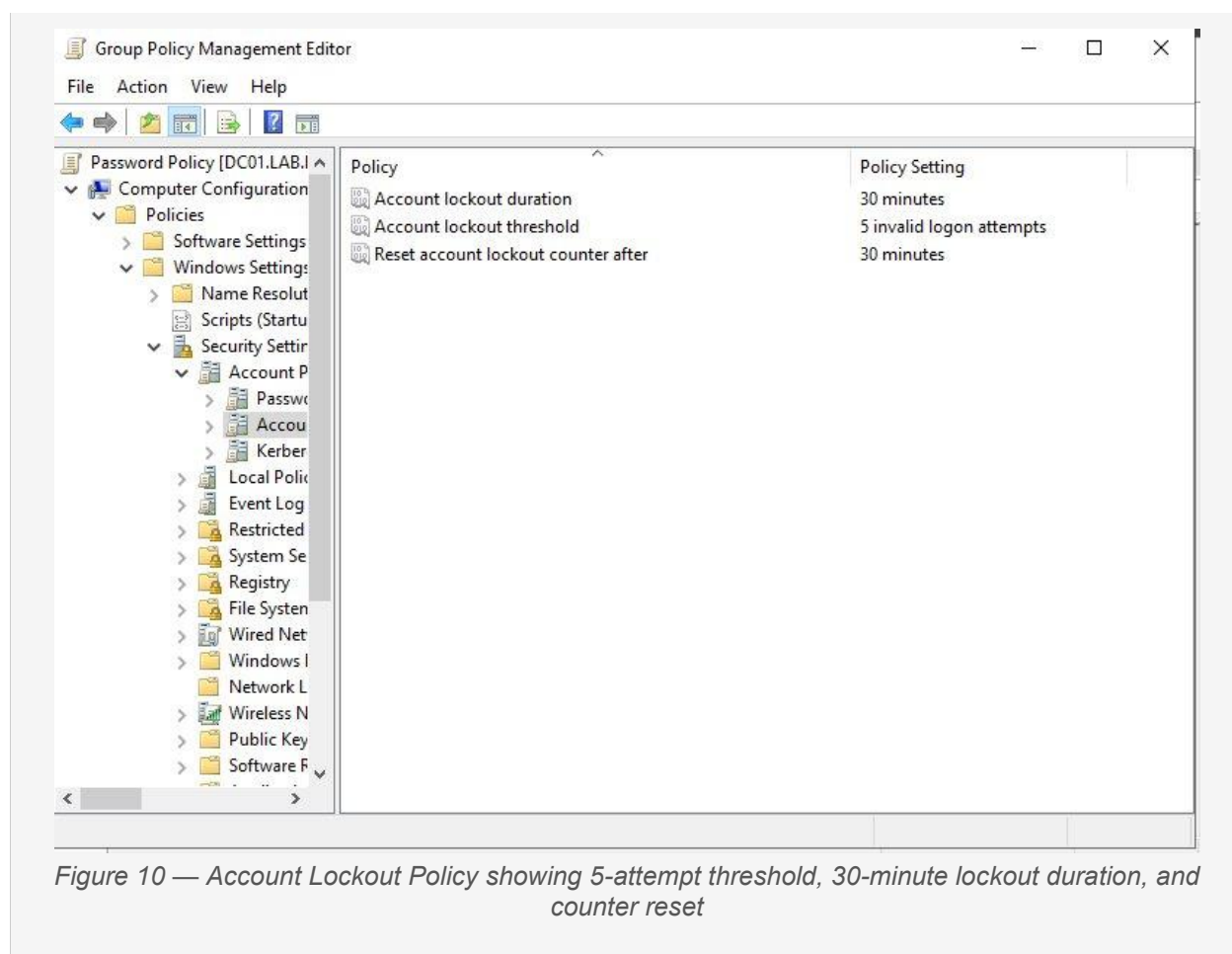


Figure 10 — Account Lockout Policy showing 5-attempt threshold, 30-minute lockout duration, and counter reset

With this lockout policy in place, a user who enters their password incorrectly 5 times will have their account locked. The Tier 1 helpdesk response is to right-click the user in ADUC → Reset Password → check "Unlock the user's account" — the same dialog demonstrated in Section 6.

Summary

This lab successfully demonstrated the core Active Directory administration skills required for a Tier 1 helpdesk role. The following tasks were completed hands-on:

1. Deployed a Windows Server domain controller with AD DS and DNS roles on lab.local
2. Created a structured OU hierarchy: Lab Users, Lab Computers, IT, and IT > HelpDesk
3. Created three domain user accounts (Jason Tran, Tina Pham, Zero Tran) with proper password policy settings
4. Created a VPN_Users security group and assigned users as members
5. Reviewed account properties including logon name, unlock option, and expiration settings
6. Performed a password reset with the "User must change password at next logon" option
7. Disabled a user account to simulate an employee offboarding workflow
8. Created and linked a Password Policy GPO to the lab.local domain via Group Policy Management Console (GPMC)
9. Configured password policy: 8-character minimum length, complexity required, 90-day maximum age
10. Configured account lockout policy: locks after 5 invalid attempts, 30-minute auto-unlock duration

These skills directly map to the most common tasks handled on a Tier 1 IT helpdesk, including new hire onboarding, password resets, account lockout resolution, employee offboarding, and enforcing security policy across the domain through Group Policy.